USER MANUAL

# BondFlux

A simulation toolkit for dynamic bonds in polymer networks

`fix bond/flux` for LAMMPS — methods, usage and validation

| | |
|---|---|
| LAMMPS style | `fix bond/flux` |
| Modes | `mc`, `3body` |
| Tested with | LAMMPS stable 30 Sep 2026 |
| Source | `https://github.com/zidan0x266/BondFlux` |

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# 1 Introduction

BondFlux simulates polymer networks whose bonds can form, break, and exchange. It is a single LAMMPS fix style, `fix bond/flux`, with two modes: Monte Carlo (MC) moves of topological bonds (`mc`), and the Sciortino three-body swap potential (`3body`). This manual explains the two methods, the LAMMPS commands, how to add the fix to LAMMPS, and the tests that validate it.

Dynamic covalent and supramolecular bonds let a polymer network change its topology while it is deformed or heated. In associative networks (vitrimers) bonds exchange partners at a constant number of cross-links [1]. In dissociative networks bonds break and re-form, so their number follows the temperature. In multiple-network elastomers, a pre-stretched filler network breaks irreversibly under load while a permanent matrix keeps the material together [2]. BondFlux handles all three in a coarse-grained model such as the Kremer–Grest bead-spring model [3]. Exchange, formation and breaking can be combined with each other and with the irreversible scission of bonds through `bond_style quartic`.

Two routes are used in the literature. In the first, a dynamic bond is an explicit LAMMPS bond, and an MC step between MD steps changes the topology (Perego and Khabaz [4], Xia et al. [5] [6], and others). In the second, a bond is a short-ranged pair attraction. A three-body repulsion enforces one bond per sticker and makes the bond swap barrier-free, so standard MD produces the exchange [7]. BondFlux provides both routes in the same fix (Table 1.1 → p. 1).

| | `mc` | `3body` |
|---|---|---|
| dynamic bond | LAMMPS bond of one type | pair interaction within $r_c$ |
| topology changes by | Metropolis MC every $N$ steps | MD (continuous forces) |
| moves | shift, swap, formation, breaking | swap, thermal breaking and formation |
| number of bonds | conserved, or fluctuating with `form/break` | fluctuating (fixed if $\varepsilon \gg kT$) |
| rate control | attempt gate $e^{-E_a/kT}$, frequency | $\lambda$, temperature |

**Table 1.1 /** The two modes of `fix bond/flux`.

# 2 Methods

## 2.1 Monte Carlo moves of topological bonds

The atoms of the fix group are *stickers*. Each sticker has at most one dynamic bond, a LAMMPS bond of the type `btype`. Every $N$ steps each sticker is visited once and offered one move (Figure 2.1 → p. 2):

- a bonded sticker $i$ with partner $k$ breaks the bond with probability $p_b$ (`break`), and otherwise attempts an exchange with a candidate $j$, chosen uniformly among the stickers within the reaction window of $i$: a *shift* $i$–$k$ + $j$ → $i$–$j$ + $k$ if $j$ is free, or a *swap* $i$–$k$ + $j$–$l$ → $i$–$j$ + $k$–$l$ if $j$ is bonded to $l$;

- a free sticker $i$ forms a bond with probability $p_f$ (`form`), with a free candidate $j$ chosen uniformly.

Every bond that is broken *or* formed must have a length in the reaction window $[r_{\min}, r_c)$. This rule makes each move reversible from the same sticker with the same proposal probability: in a shift, the free stickers near $i$ before the move are the free stickers near $i$ after it, with $j$ and $k$ exchanged, and both are inside the window. Candidates must be bondable (`bondable` types, `molecule` rule) and must not already be bonded to $i$ through another bond.

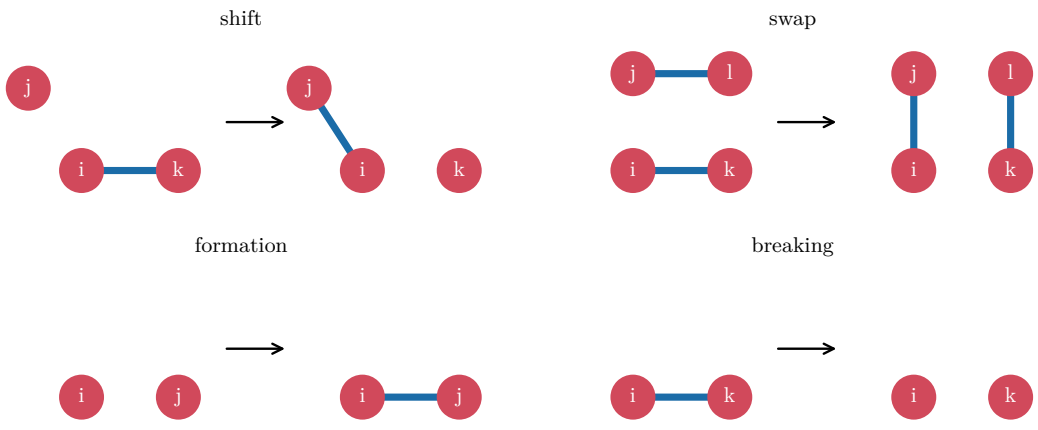



**Figure 2.1 /** Monte Carlo moves of `fix bond/flux mc`: sticker $i$ is the sticker offered the move; dynamic bonds are drawn as lines. Top: exchange (shift, swap), which keeps the number of dynamic bonds. Bottom: formation and breaking, which change it.

## Energy change

The energy change of a move is evaluated with the bond and pair styles of the simulation:

$$\Delta E = \sum_{\text{formed}} \Big[U_{\text{bond}}(r) - \Delta U_{\text{pair}}(r)\Big] - \sum_{\text{broken}} \Big[U_{\text{bond}}(r) - \Delta U_{\text{pair}}(r)\Big] + E_{\text{bond}}\,(n_{\text{formed}} - n_{\text{broken}}), \tag{2.2}$$

where $\Delta U_{\text{pair}}(r) = U_{\text{pair}}(r;1) - U_{\text{pair}}(r;w_{12})$ is the change of the pair energy when a pair stops being a 1–2 neighbor ($w_{12}$ = `special_bonds` weight) and $E_{\text{bond}}$ is an optional association energy (`ebond`). For a Kremer–Grest model with `special_bonds lj 0 1 1`, a FENE bond and a WCA pair, the WCA terms cancel and $\Delta E$ is the change of the FENE energy; with `special_bonds 1 1 1` (required by `bond_style quartic`) $\Delta U_{\text{pair}} = 0$. $\Delta E$ is exactly the change of the LAMMPS potential energy, as checked in Chapter 4 → p. 11.

## Acceptance

Exchange moves are accepted with $\min(1, e^{-\Delta E/kT})$. Formation and breaking are each other's reverse but are proposed with different probabilities (a free sticker picks one of $n_c$ candidates; a bonded sticker breaks with probability $p_b$), so they carry the Metropolis–Hastings ratio

$$P_{\text{form}} = \min\!\left(1, \frac{p_b\, n_c}{p_f}\; e^{-\Delta E/kT}\right), \qquad P_{\text{break}} = \min\!\left(1, \frac{p_f}{p_b\, n_c'}\; e^{-\Delta E/kT}\right), \tag{2.3}$$

where $n_c'$ is the number of free candidates of $i$ after the bond is broken.

## Barrier

A reaction barrier $E_a$ (`barrier`) and an attempt probability $P$ (`prob`) enter as a gate: a sticker is offered a move only with probability $g = P\, e^{-E_a/kT}$. The gate multiplies the rate of every move and of its reverse by the same factor, so it changes the kinetics and leaves the equilibrium unchanged (Section 4.2 → p. 12). Adding $E_a$ to $\Delta E$ inside the Metropolis exponent, as in Ref. [6], does not have this property.

## Parallel moves

A move centered on sticker $i$ involves atoms within $2r_c$ of $i$. Each MPI sub-domain is split into $2^d$ sectors, and the sectors are processed one after the other; moves in the same sector of neighboring processors are then at least half a sub-domain apart and cannot conflict. Partner changes of ghost atoms are sent to their owners after each sector, also through the multi-hop paths near sub-domain corners. At the end of an MC step the bond and 1–2 special lists are updated and a reneighboring is triggered. The partner bookkeeping is rebuilt from the bond list at every MC step, so bonds broken by `bond_style quartic` are picked up automatically.

### Contact diagnostics

A sticker is *in contact* while it has a candidate for its move: an exchange candidate if it is bonded (with its bond inside the window), a free candidate if it is free and bonds can form. Contacts open and close with the motion of the stickers, while moves are only offered every $N$ steps. With `contact` $N_c$ ($N_c$ a divisor of $N$) the fix checks the contacts every $N_c$ steps and reports the time in contact summed over the stickers, the completed contact episodes, the episodes during which no MC step took place, their fraction $f_{\mathrm{miss}}$, and $k_{\mathrm{cond}}$, the accepted exchanges and formations per unit contact time. A large $f_{\mathrm{miss}}$ means that the MC interval filters out short contacts, so that the exchange rate depends on $N$ in a way that the barrier does not describe; $N$ should then be reduced.

## 2.2 Three-body swap potential

In `3body` mode the fix adds the Sciortino potential [7] between bondable stickers. The two-body bond is a $2n$–$n$ Lennard-Jones potential,

$$V(r) = \varepsilon\,\frac{\phi(r)-\phi(r_c)}{1+\phi(r_c)}, \qquad \phi(r) = 4\Big[\Big(\frac{\sigma}{r}\Big)^{2n} - \Big(\frac{\sigma}{r}\Big)^{n}\Big], \qquad r < r_c, \tag{2.4}$$

shifted to zero at $r_c$ and rescaled to a depth of $-\varepsilon$ at $r_{\min} = 2^{1/n}\sigma$. With `eshift no`, $V = \varepsilon\,\phi(r)$ is used unshifted, as in Ref. [7]. The three-body term,

$$E_3 = \lambda\,\varepsilon \sum_i \sum_{j<k} V_3(r_{ij})\,V_3(r_{ik}), \qquad V_3(r) = \begin{cases} 1 & r \le r_{\min} \\ -V(r)/\varepsilon & r_{\min} < r < r_c \end{cases}, \tag{2.5}$$

penalizes a sticker with two partners. For $\lambda = 1$ the energy gained by a second partner is compensated exactly, so the swap is barrier-free; $\lambda > 1$ adds an exchange barrier of $(\lambda-1)\varepsilon$. Since $V_3$ is built from $V$ itself, this holds for any bond shape. Forces, energy and virial are computed in the `pre_reverse` stage on a full neighbor list, and forces on ghost atoms are summed by LAMMPS's reverse communication. The fix supplies the complete interaction between bondable sticker types; the pair style interaction between them must be switched off (`pair_coeff A B 0.0 ...`).

### Stillinger–Weber shaped bond

With `3body sw` the bond has the Stillinger–Weber form

$$V(r) = \frac{\varepsilon}{N}\left[B\Big(\frac{\sigma}{r}\Big)^{p} - \Big(\frac{\sigma}{r}\Big)^{q}\right]\exp\Big(\frac{\sigma}{r-r_c}\Big), \qquad r < r_c, \quad p > q \ge 0, \tag{2.6}$$

which goes to zero with all its derivatives at $r_c$, so neither the energy nor the force jumps at the cutoff. The minimum $r_{\min}$ is the first zero of $\mathrm{d}V/\mathrm{d}r$ below $r_c$, found by bisection to machine precision, and $N$ sets $V(r_{\min}) = -\varepsilon$.

## Several sticker chemistries

The fix arguments are the parameters of every bondable pair of types; the `coeff` keyword sets them, e.g. $\varepsilon_{ab}$, $\sigma_{ab}$, $n_{ab}$, $r_{c,ab}$ and $\lambda_{ab}$, for selected pairs $a$–$b$. Each bond, and each leg of the three-body term, uses the parameters of its own pair of types. The three-body term of a triplet $j$–$i$–$k$ is

$$E_3^{(jik)} = \min\Big(\lambda_{ij}\varepsilon_{ij},\ \lambda_{ik}\varepsilon_{ik}\Big)\, V_3^{(ij)}(r_{ij})\, V_3^{(ik)}(r_{ik}). \tag{2.7}$$

The minimum generalizes Sciortino's compensation: with $\lambda = 1$, a partner $k$ approaching a stronger bond $i$–$j$ leaves the energy at $-\varepsilon_{ij}$, and a stronger partner approaching a weaker bond lowers the energy monotonically to the stronger bond, so a swap between bonds of different strength has no barrier beyond their energy difference. For equal pairs the expression reduces to the single-chemistry potential. The `triplet` keyword replaces the minimum by a chosen strength for the triplets centered on one type with legs of two given types.

The three-body potential is widely used for coarse-grained vitrimers, for example by Liu, Ganesan and co-workers to relate the bond swap barrier to the structure, mechanics, self-healing and reprocessing of vitrimer networks [8], and by Sing and co-workers to study penetrant diffusion in vitrimer networks [9].

# 3 Using fix bond/flux in LAMMPS

## 3.1 Syntax

```
fix ID group-ID bond/flux mc Nevery btype rc T seed &
    keyword value ...
fix ID group-ID bond/flux 3body [lj] eps sigma n rc lambda &
    keyword value ...
fix ID group-ID bond/flux 3body sw eps sigma B p q rc lambda &
    keyword value ...
coeff values:   itype jtype (the values of the bond shape)
triplet values: itype jtype ktype strength
```

- `mc`: `Nevery` = steps between MC steps; `btype` = bond type of the dynamic bonds; `rc` = upper end of the reaction window (distance); `T` = temperature of the Metropolis test (use the thermostat set point); `seed` = random number seed.
- `3body`: `eps`, `sigma`, `n` = depth, size and exponent of the $2n$–$n$ bond (`lj`, the default); `B`, `p`, `q` = parameters of the Stillinger–Weber shaped bond (`sw`); `rc` = cutoff; `lambda` = three-body strength.

### Output

`mc`: global vector with 16 elements (attempts, proposals, accepted shifts, swaps, formations, breakings, geometric rejections, Metropolis rejections, sum of $\Delta E$ of accepted moves, number of dynamic bonds, accepted moves in the last MC step; with `contact`: contact time, contact episodes, missed episodes, $k_{\mathrm{cond}}$, $f_{\mathrm{miss}}$); global scalar $E_{\mathrm{bond}}\times$(number of dynamic bonds) when `ebond` is set, included in the potential energy. `3body`: global scalar = $E_2 + E_3$ (included in the potential energy and virial), vector ($E_2$, $E_3$, bonded pairs, stickers with more than one partner). Per-atom array, both modes: column 1 = partner atom ID (nearest bonded partner in `3body`), column 2 = number of moves of the atom (`mc`) or number of partners within `rbond` (`3body`).

### Restrictions

`mc`: molecular atom style; orthogonal box; `special_bonds lj/coul` $w_{12}$ `1.0 1.0` (1–3 and 1–4 neighbors must not be special); `comm_modify cutoff` $\geq 2r_c +$ skin$/2$; processor sub-domains $\geq 8r_c +$ skin; `extra/bond/per/atom` and `extra/special/per/atom` when bonds can form; angles or dihedrals must not contain dynamic bonds. For FENE bonds $r_c < 0.95R_0$; for quartic bonds $r_c$ must be shorter than the breaking distance. LAMMPS cannot combine `quartic` with unbreakable styles

| keyword | mode | value | default |
|---|---|---|---|
| `bondable` | both | `itype jtype` (ranges allowed, repeatable) | all pairs |
| `shift`, `swap` | mc | `yes`/`no` | `yes` |
| `form`, `break` | mc | attempt probabilities $p_f$, $p_b$ (both $> 0$ or both 0) | 0, 0 |
| `ebond` | mc | association energy per dynamic bond | 0 |
| `barrier` | mc | activation energy $E_a$ | 0 |
| `prob` | mc | attempt probability $P$ | 1 |
| `rmin` | mc | lower end of the reaction window | 0 |
| `molecule` | mc | `any`, `inter`, `intra` | `any` |
| `contact` | mc | contact diagnostics every $N_c$ steps ($N_c$ divides $N$) | off |
| `coeff` | 3body | parameters of a pair of bondable types (repeatable) | fix args |
| `triplet` | 3body | three-body strength $\lambda\varepsilon$ of a triplet (repeatable) | min rule |
| `rbond` | 3body | distance that defines a bond for the output | `rc` of the pair |
| `eshift` | 3body | `yes`/`no`: shift $V$ to zero at `rc` | `yes` |

**Table 3.1 /** Keywords of `fix bond/flux`.

in `bond_style hybrid`, so a network with breakable quartic strands uses quartic dynamic bonds as well. `3body`: `newton on`; `comm_modify cutoff` $\geq r_c +$ skin (largest $r_c$ of all pairs). Neither mode supports `run_style respa`. The fix writes no restart data: after `read_restart` it is defined again, the partners are rebuilt from the bond list, and the per-atom move counts and random number streams start anew.

## 3.2 Examples

### Kremer–Grest vitrimer

Exchange only, number of bonds conserved (`examples/kg_network`):

```
special_bonds   lj/coul 0.0 1.0 1.0
bond_style      fene
bond_coeff      * 30.0 1.5 1.0 1.0
group           stick type 2
fix             flux stick bond/flux mc 100 2 1.3 1.0 777 barrier 2.0
```

### Dissociative network

Reversible formation and breaking with association energy:

```
fix             flux stick bond/flux mc 100 2 1.3 1.0 12347 &
                form 1.0 break 0.5 ebond -24.0
```

### Multiple network with breakable strands

Quartic bonds; type 1 = permanent strands, type 2 = dynamic junctions; strands break irreversibly above $1.3\sigma$:

```
special_bonds   lj/coul 1.0 1.0 1.0
bond_style      quartic
bond_coeff      1 1200.0 -0.55 0.25 1.3 34.6878
bond_coeff      2 1200.0 -0.55 0.25 1.3 34.6878
fix             flux stick bond/flux mc 10 2 1.2 1.0 4321 barrier 3.0
```

### Cross-linker vitrimer

As in Refs. [5] [6]: A stickers on chains, B ends of three-bead cross-linkers, LJ with $r_c = 1.5\sigma$, MC every 10 steps, reaction window $0.87 \leq r < 1.14$; only free A stickers attack bonded B:

```
special_bonds   lj/coul 0.0 1.0 1.0
```

```
bond_coeff      2 30.0 1.5 0.0 1.0      # FENE without the WCA term
fix             flux AB bond/flux mc 10 2 1.14 1.0 9876 &
                bondable 1 2 swap no rmin 0.87 barrier ${Ub}
```

### Pair-swap vitrimer

As in Ref. [4]: all stickers bonded, swaps only, an MC step every MD step:

```
fix             flux stick bond/flux mc 1 2 1.3 1.0 5555 shift no
```

### Sciortino dumbbells

`examples/dumbbells`; A–A and B–B repulsive $2n$–$n$ WCA:

```
pair_style      mie/cut $(2.0^(1.0/100))
pair_coeff      * * 1.0 1.0 200 100
pair_coeff      1 2 0.0 1.0 200 100
pair_modify     shift yes
fix             tb stick bond/flux 3body 1.0 1.0 100 1.3 1.0 &
                bondable 1 2
```

The model of Ref. [9] corresponds to `3body 30.0 1.0 100 1.2 lambda`.

### Two sticker chemistries

A stickers (type 1) bond to B (type 2) with $\varepsilon = 1$ and to C (type 3) with $\varepsilon = 0.6$:

```
fix             tb stick bond/flux 3body 1.0 1.0 100 1.3 1.0 &
                bondable 1 2 bondable 1 3 coeff 1 3 0.6 1.0 100 1.3
    1.0
```

## 3.3 Adding BondFlux to LAMMPS

The fix uses only core LAMMPS classes and is compiled as part of the LAMMPS core. With CMake:

```
cd lammps/src
# link (or copy) the two files
ln -s ../../src/fix_bond_flux.h ../../src/fix_bond_flux.cpp .
cd ..
cmake -S cmake -B build -D PKG_MOLECULE=on -D PKG_EXTRA-PAIR=on \
      -D BUILD_MPI=on -D BUILD_SHARED_LIBS=on
cmake --build build -j 8
build/lmp -h | grep bond/flux
```

`PKG_MOLECULE` provides the bond styles, `PKG_EXTRA-PAIR` the `mie/cut` pair style of the dumbbell examples, and `BUILD_SHARED_LIBS` the library for the Python tests. With the legacy make build, copy the two files into `lammps/src` and run `make mpi`. BondFlux was developed and tested with LAMMPS stable 30 Sep 2026.

# 4 Validity tests

The scripts are in `tests/` and `examples/`; the figure data are in `doc/data/bondflux_validation.h5`. Times are in Lennard-Jones units $\tau = \sigma\sqrt{m/\varepsilon}$. The bond autocorrelation $C(t)$ is the fraction of bonds present at time 0 that are present at time $t$; $\tau_x$ is its decay time.

## 4.1 Exact detailed balance of the MC moves

Six stickers are frozen in place (no time integration), so only the MC moves change the system. The visited topologies must then follow $e^{-E/kT}$ over all topologies the moves can reach. The reachable topologies and their energies are enumerated independently of LAMMPS (`tests/test_mc_balance.py`). The test was run for four force fields: Kremer–Grest FENE with `special_bonds 0 1 1`; quartic with `1 1 1`; FENE without WCA, where bonded pairs lose their repulsion; and harmonic with a 1–2 weight of 0.5. It covered exchange only, formation and breaking only, and both together, with and without a barrier, on 1, 2 and 4 MPI ranks, with the cluster in the middle of the box and across a periodic box corner. All 48 runs pass: the largest block z-score is below 3.1, the total variation distance below 0.006, and the LAMMPS potential energy changes by the sum of the $\Delta E$ of the accepted moves to $10^{-10}$. Figure 4.1 →p.11 shows the run with formation and breaking on 4 ranks.

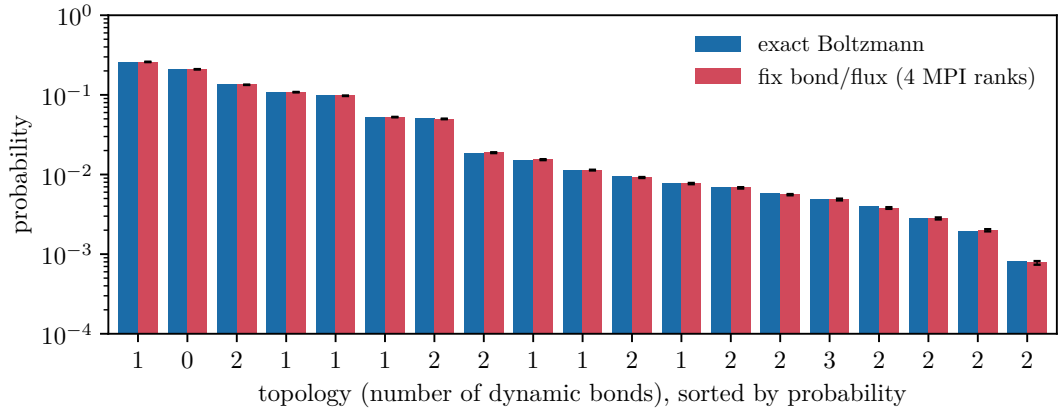



**Figure 4.1 /** Probabilities of the 19 reachable topologies of a frozen six-sticker cluster (0, 1 or 2 dynamic bonds) with exchange, formation and breaking ($p_f = p_b = 0.5$, $E_{\mathrm{bond}} = -20\varepsilon$, $kT = 4\varepsilon$, Kremer–Grest FENE, $4 \times 10^5$ MC steps on 4 MPI ranks) compared with the exact Boltzmann distribution. Total variation distance 0.0018.

## 4.2 Barrier as a gate versus inside the Metropolis exponent

For the same cluster, the transition matrix of the exchange MC was built exactly and its stationary distribution compared with the Boltzmann distribution (`tests/compare_acceptance_rules.py`). With the gate of `fix bond/flux` the stationary distribution is Boltzmann for every barrier. With the barrier added to $\Delta E$, a move and its reverse are both penalized, so for $|\Delta E| < E_a$ their acceptance ratio becomes $e^{-2\Delta E/kT}$. The equilibrium of the topology is then that of half the temperature (Table 4.2 → p. 12).

| rule | window | $E_a/kT = 0$ | 1 | 2 | 4 | 8 |
|---|---|---|---|---|---|---|
| gate | $[0, 1.3)$ | $2 \times 10^{-15}$ | $2 \times 10^{-14}$ | $1 \times 10^{-14}$ | $1 \times 10^{-13}$ | $2 \times 10^{-11}$ |
| exponent | $[0, 1.3)$ | $2 \times 10^{-15}$ | 0.235 | 0.275 | 0.283 | 0.283 |
| gate | $[0.87, 1.14)$ | $3 \times 10^{-17}$ | $6 \times 10^{-16}$ | $2 \times 10^{-15}$ | $1 \times 10^{-14}$ | $4 \times 10^{-13}$ |
| exponent | $[0.87, 1.14)$ | $3 \times 10^{-17}$ | 0.149 | 0.149 | 0.149 | 0.149 |

**Table 4.2 /** Total variation distance between the stationary distribution and the Boltzmann distribution (exact, no sampling). Gate (`fix bond/flux`): $P_{\text{acc}} = e^{-E_a/kT} \min(1, e^{-\Delta E/kT})$; exponent: $P_{\text{acc}} = \min(1, e^{-(\Delta E + E_a)/kT})$.

A reaction window checked only for the new bond [5] is not reversible either: a bond stretched beyond the window can be exchanged away but never formed. This bias acts through bond-length fluctuations and does not appear for a frozen cluster.

## 4.3 Kremer–Grest sticker network

A melt of 200 Kremer–Grest chains of 20 beads with stickers at beads 2, 6, 10, 14 and 18 (`examples/kg_network`) is first turned into a dissociative network: stickers form and break bonds with $E_{\text{bond}} = -24\varepsilon$ at $kT = 1$, and the number of bonds settles at about 400 of 500 possible (Figure 4.3 (a)). With formation and breaking switched off, the number of bonds is conserved (409 bonds before and after $2 \times 10^5$ steps for every barrier) and the shift and swap moves relax the topology. The barrier enters as the gate $e^{-E_a/kT}$: going from $E_a = 0$ to $2\varepsilon$ and $4\varepsilon$ reduces the number of attempts by factors of 7.40 and 7.33 per step of $2kT$ and the number of accepted exchanges by 7.45 and 7.2, close to $e^2 = 7.39$. The bond autocorrelation time grows from $\tau_x = 128\tau$ to $739\tau$ and about $4300\tau$ (the last value extrapolated beyond the run), a factor of 5–6 per $2kT$, because exchanges that swap a bond back restore the original pairing (Figure 4.3 (b)).

With `contact 10` and $E_a = 0$, MC steps every 100 steps (as in the example) miss 42 % of the contact episodes resolved every 10 steps, and $k_{\text{cond}}$ is $0.012\,\tau^{-1}$; with MC steps every 10 steps no episode is missed and $k_{\text{cond}} = 0.115\,\tau^{-1}$. The barrier

scaling above is not affected, but rates in this example are set partly by the MC interval.

## 4.4 Contact diagnostics

`tests/test_contact.py` runs a gas of A–B dimers and free A stickers in which the moves are blocked by a barrier of $60\,kT$, so that the topology stays fixed and the contacts can be recounted exactly from the trajectory written at every check. The contact time, the number of episodes and the number of missed episodes equal the recount on 1, 2 and 4 ranks, for MC steps at every check (no missed episodes) and at every tenth check.

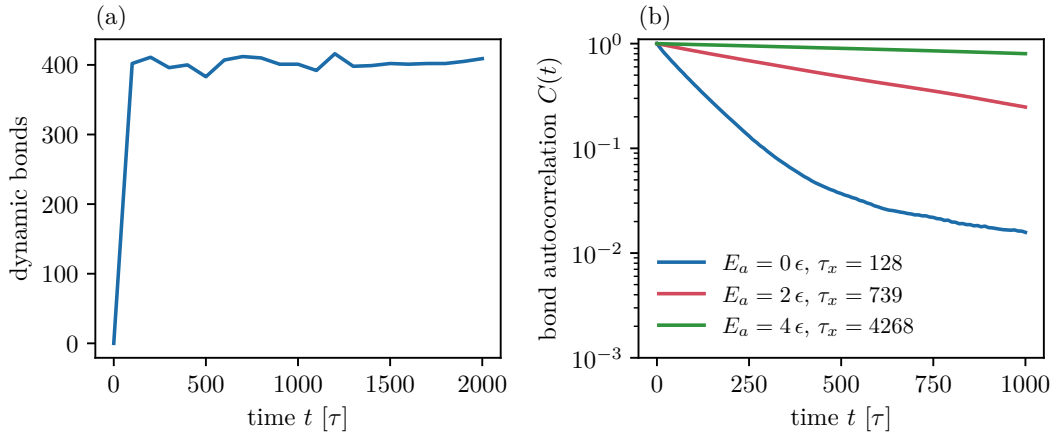



**Figure 4.3 /** Kremer–Grest sticker network. (a) Number of dynamic bonds during reversible formation and breaking. (b) Bond autocorrelation $C(t)$ in the same network with exchange only, for three barriers.

## 4.5 Three-body forces and virial

`tests/test_3body.py` compares the forces with central finite differences of the total energy, and the virial with finite-difference strain derivatives (box scaling in $x$, $y$, $z$ and an $xy$ shear). The test uses $n = 12$ and 100, $\lambda = 1$ and 1.3, and 1, 2 and 4 MPI ranks. The relative errors are $\leq 8 \times 10^{-10}$ for the forces and $\leq 3 \times 10^{-9}$ for the virial, and the parallel forces agree with the serial ones to $2 \times 10^{-13}$. For an A–B bond at $r_{\min}$ and a second B approaching from the other side, the total energy stays $-\varepsilon$ to $10^{-12}$ while the two-body energy varies between $-2\varepsilon$ and $-\varepsilon$ ($n = 100$, $\lambda = 1$).

With `--coeff`, C stickers bond to A with their own parameters ($\varepsilon = 1.6$, $\sigma = 0.95$, $n = 18$, $r_c = 1.25$, $\lambda = 1.2$), so the cluster contains triplets with one B and one C leg. Forces and virial agree with finite differences to the same accuracy on 1, 2 and 4 ranks, and a `coeff` equal to the default parameters reproduces the energy and forces exactly. For the mixing rule, A–B ($\varepsilon = 1$) and A–C ($\varepsilon = 0.6$) bonds with $n = 100$ and $\lambda = 1$ are used: a C approaching an A–B bond leaves the energy at

$-1$ to $10^{-12}$, and a B approaching an A–C bond lowers it monotonically, following $E = -0.6 - 0.4\,V_3^{(AB)}$ to $10^{-16}$. A `triplet` strength of 1.0 for A with B and C legs replaces the minimum: the energy then follows $E = -1 + 0.4\,V_3^{(AC)}$ to $10^{-16}$, a barrier of $0.4\varepsilon$.

With `--form sw` the same checks are made for the Stillinger–Weber shaped bond ($B = 0.5$, $p = 12$, $q = 6$, $r_c = 1.5$; and $\varepsilon = 1.6$, $\sigma = 0.95$, $B = 0.6$, $p = 10$, $q = 4$, $r_c = 1.4$, $\lambda = 1.2$ for A–C), with a triplet strength of 0.8 for A with B and C legs: forces and virial agree with finite differences to the same accuracy on 1, 2 and 4 ranks. At the minimum found independently in the test, the bond energy is $-\varepsilon$ to $10^{-12}$, and the landscape for $\lambda = 1$ is flat to $10^{-12}$.

## 4.6 Dumbbells: comparison to the reference and bond-swap tuning

The dumbbell system of Ref. [7] has 400 A and 600 B particles at $\rho = 0.125$, with $n = 100$, $r_c = 1.3$ and $kT = 0.03$ (`examples/dumbbells/in.dumbbells`: $1000\tau$ equilibration, $2000\tau$ production, Langevin damping time 100). All 400 A particles are bonded at all times, and for $\lambda = 1$ the bond autocorrelation decays exponentially with $\tau_x = 116.7\tau$, against $116\tau$ in Ref. [7] (Figure 4.4 (a)). For $\lambda = 1.1$ the lifetime grows 3.5 times to $\tau_x = 404\tau$, and for $\lambda = 1.3$ the bonds practically do not exchange within $1000\tau$.

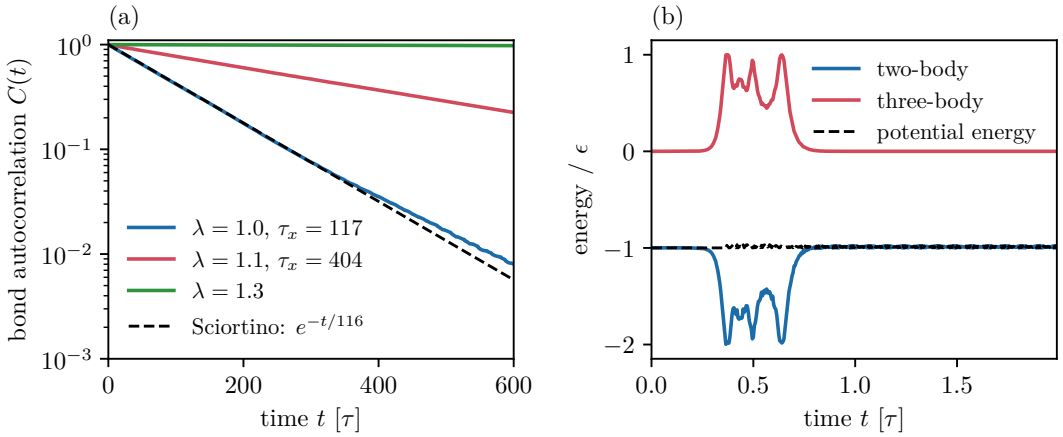



**Figure 4.4 /** Three-body swap potential. (a) Bond autocorrelation $C(t)$ of the dumbbell system for $\lambda = 1$, 1.1 and 1.3, with the reference decay $e^{-t/116}$ of Ref. [7]. (b) Energies during a burst of bond swaps of one A and two B particles ($n = 150$, $\lambda = 1$, NVE, $kT \approx 0.01$); the two-body and three-body energies compensate and the total potential energy stays at $-\varepsilon$.

## 4.7 Potential energy during a swap event

One A and two B particles move in NVE in a box of side 2.6 ($n = 150$, $\lambda = 1$, `examples/dumbbells/in.swap_event`). During a swap, the two-body energy drops to $-2\varepsilon$ while the three-body energy rises to $+\varepsilon$, so the potential energy stays at $-\varepsilon$ (Figure 4.4 (b)). The total energy is conserved to $10^{-5}$ over $1.5 \times 10^5$ steps.

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